Lecture 8 · Part II · Lecture notes

PyTorch

Dataset: Fashion-MNIST

How to use these notes

The primary text for this lecture is Géron, Chapter 10. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

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1 Where we are

Lecture 7 found four things `MLPClassifier` will not let you do: change what it optimises, see a gradient, stop mid-epoch, or move the computation to another device. All four are one wall — `fit()` contains a loop over epochs and batches that you did not write, cannot see and cannot enter.

The repair is not a better library. It is *owning the loop*. Today: what that loop has to compute, why it can be computed at all, and how to write it.

2 Derivation: backpropagation as reverse-mode autodiff

You called `clf.fit()`. It computed 266,610 partial derivatives, on every batch, twenty times over. *How?* The cost of that computation is what decides whether training a network is possible at all — and the reason it is possible is a property of the *direction* in which the chain rule is applied.

2.1 Two bracketings, two algorithms

A network is a composition, so writing $L = f_3(f_2(f_1(\boldsymbol{\theta})))$ gives

$$\frac{\partial L}{\partial \boldsymbol{\theta}} = J_3 J_2 J_1,$$

with J_i the Jacobian of f_i evaluated at the point the forward pass reached. Every layer contributes one factor — and matrix multiplication is associative:

$$\underbrace{(J_3 J_2) J_1}_{\text{reverse mode}} \;=\; \underbrace{J_3 (J_2 J_1)}_{\text{forward mode}} .$$

Same number. Wildly different cost.

Forward mode picks a direction $\mathbf{v}$ in input space and carries a tangent $\dot{u} = \partial u / \partial v$ alongside every intermediate value. One sweep gives the derivative of every output with respect to *one* input direction, and nothing needs to be stored, because the tangent travels with the value.

Reverse mode picks a direction in output space — for a scalar loss there is only one, and it is 1 — and carries an adjoint $\bar{u} = \partial L/\partial u$ backwards. One sweep gives the derivative of *one* output with respect to every input, and the forward values must be kept, because the Jacobians are evaluated at them.

Mode	Sweeps needed	Cheap when
Forward	n — one per input	few inputs, many outputs
Reverse	m — one per output	many inputs, few outputs

2.2 A worked example small enough to check

$$L = w^2, \qquad w = xy + \sin x, \qquad \text{at } x = 2,\ y = 3.$$

By hand, $\partial L/\partial x = 2w(y + \cos x)$ and $\partial L/\partial y = 2wx$.

Node	Expression	Value	Adjoint	Computed as	Value
x	input	2.0	$\partial L/\partial L$	the seed	1
y	input	3.0	$\partial L/\partial w$	$2w$	13.8186
u	$x \cdot y$	6.0000	$\partial L/\partial u$	$(\partial L/\partial w) \cdot 1$	13.8186
s	$\sin x$	0.9093	$\partial L/\partial s$	$(\partial L/\partial w) \cdot 1$	13.8186
w	$u + s$	6.9093	$\partial L/\partial y$	$(\partial L/\partial u)\, x$	27.6372
L	w^2	47.7384	$\partial L/\partial x$	$(\partial L/\partial u)\, y + (\partial L/\partial s) \cos x$	35.7052

Six nodes evaluated once each in topological order, then six lines backwards — one pass, both derivatives. Forward mode would need two sweeps here, one seeded at x and one at y.

The only bookkeeping in the whole method

x is used twice, once by the product and once by the sine, so

$$\frac{\partial L}{\partial x} = \underbrace{\frac{\partial L}{\partial u}\, y}_{\text{via } \times} + \underbrace{\frac{\partial L}{\partial s} \cos x}_{\text{via sin}}.$$

A node's adjoint is the *sum* of the contributions along its outgoing edges. Miss one edge and the gradient is silently wrong, not absent.
Reverse-mode autodiff is a topological sort, a table of local derivatives, and this sum. There is no other idea in it.

The notebook checks this against `torch`: autograd returns the same two numbers as the hand derivatives, to machine precision.

2.3 The argument that settles it

Training differentiates a function from 266,610 parameters to *one* scalar loss. Forward mode needs one sweep per input: 266,610 of them. Reverse mode needs one per output: 1.

Reverse mode is not a good choice

It is the *only viable* one, by a factor equal to the parameter count.

So what is backpropagation?

Reverse-mode automatic differentiation, applied to a neural network, with the seed set to 1 at the loss. It is not specific to neural networks, it is not an approximation, and it is not gradient descent — it supplies the gradient that descent then uses. The 1986 paper's contribution was noticing this was cheap enough to be practical, not inventing the chain rule.

3 What the wrapper costs

The accuracy was not the bug. 87.38% validation accuracy against a 10.0% baseline is a real result: the split was clean, the validation set separate, the test set touched once. The bug is that you cannot do anything else.

A cost in time has to be read in pairs. Scikit-Learn and PyTorch on the same CPU isolate the framework; the same PyTorch code on the CPU and on the accelerator isolates the device. Timing PyTorch on the accelerator against Scikit-Learn on the CPU would give one number for two changes.

3.1 So when does the accelerator pay?

A forward pass is $\phi(\mathbf{XW} + \mathbf{b})$ repeated, and a GPU is a machine for large matrix products and very little else. Ours are not large: 266,610 parameters and a batch of 128 is a small matrix product, so much of each step is the cost of dispatching it rather than the arithmetic.

Widen the layers and measure again, with identical code on both devices. At the smallest width the two devices nearly tie; from there the ratio climbs to the widest, because each batch carries more arithmetic to cover the cost of dispatching it. Measured with 2 epochs on 12,000 images, batch 128, identical code on both devices, on an Apple MPS backend; a Colab T4 has a different crossover, in the same place in the argument.

4 Tensors, autograd, and the loop

A tensor is an array that also knows two things a NumPy array does not: where it lives — an operation between tensors on different devices is an error, and a loud one — and whether to record.

`requires_grad` does not compute a derivative. It says *record what happens to me*. Every operation on a recording tensor creates a node holding the operation and its inputs; the graph is built during the forward pass, by running it, with no separate compilation step; and it is rebuilt from scratch on every batch, which is why a Python `if` inside `forward` just works. That is the forward sweep from §2, with the graph written down as it goes so the reverse sweep has something to walk — and you can inspect it: `r.grad_fn` and `r.grad_fn.next_functions` are the nodes of the worked example,

by name.

Evaluation needs no derivatives, so `torch.no_grad()` stops the recording and saves memory proportional to the batch. Forgetting it is slow and memory-hungry, but it is not a correctness bug. The two that follow are.

```
for epoch in range(EPOCHS):
    model.train()
    for xb, yb in train_loader:
        opt.zero_grad()          # 1  clears p.grad, which backward() adds to
        out  = model(xb)         # 2  the forward sweep; the graph is recorded
        loss = lossf(out, yb)    # 3  reduces the batch to one scalar
        loss.backward()          # 4  the reverse sweep; fills p.grad
        opt.step()               # 5  updates the parameters from p.grad
```

Five lines. There is no hidden machinery: this is the whole of what `fit()` was doing — and every arrow is a line you wrote, so every arrow is a place you can print, clip, log, branch or stop.

Rebuilt in PyTorch and run on MPS, the model reaches 86.32% validation accuracy against Scikit-Learn's 87.38%. The accuracies agree, as they must: same architecture, same optimiser, same learning rate, same epochs, same data. *The rebuild is not an improvement — it is the same model.* What we bought is the loop. Overlaying the two training-loss curves is the first check after any rewrite; if they had disagreed, one of the two implementations would be wrong.

5 Two bugs that never raise

5.1 The missing `zero_grad()`

`backward()` *accumulates*: `p.grad` is added to, not overwritten, so leaving out the line that clears it is not an error.

Failure condition — what it does to training

At step t the optimiser sees the sum of every gradient computed so far, not the current one. The effective step size grows without bound, and Adam's normalisation hides the divergence for a while. No exception, no warning, no `NaN`.

	Final training loss	Validation accuracy
With `opt.zero_grad()`	0.3082	86.32%
Without it	2.2441	20.44%

65.88 points: 10 epochs on 12,000 images, identical seeds, identical everything else.

5.2 The missing `model.eval()`

Some layers behave differently in the two phases. Dropout zeroes a random fraction of units in `train()` and passes everything through in `eval()`; batch normalisation uses this batch's statistics in one and the running averages in the other. *The module defaults to training mode*, so building

a model and never calling `eval()` means every evaluation you make is of a randomly crippled network. Batch normalisation arrives in Lecture 9; the same failure applies to it and is harder to see, because it does not fluctuate.

Evaluation mode	Test accuracy
`model.eval()`	85.54%
`model.train()`, mean of 10 calls	84.75%
lowest of those 10	84.53%
highest of those 10	84.97%

Dropout 0.2, measured on 10,000 test images: the missing call costs 0.79 points.

The tell is the spread, not the mean

0.44 accuracy points across ten *identical* calls. A deterministic function of fixed weights and fixed data returns the same number every time, so if your metric wobbles between calls, ask which layer is still in training mode.
Run any evaluation twice. It costs one line, and it catches this bug, the seeding bugs, and any accidental shuffling of the labels.

Both bugs are one line, neither raises, neither warns, and both produce a plausible number — costing 65.88 points and 0.79 points respectively. One is caught by reading the loop, the other by running it twice, and both survive a review that is looking for exceptions rather than for behaviour.

6 Feeding the loop, and one more silent error

`Dataset` answers two questions — how many, and give me number i — and `DataLoader` shuffles, batches, and can load in parallel while the device is busy. Indexing a big tensor by hand works while the data fits in memory; it stops working once the data does not, and the interface does not change.

Failure condition — the last batch is short

10,000 test images in batches of 384 gives 27 batches, the last containing 16 — so a plain mean of per-batch accuracies weights those 16 exactly as heavily as a full batch.

How the accuracy was aggregated	Value
Counted over the whole set	85.350%
Mean of the per-batch accuracies	85.204%

0.15 points, and this time the per-batch mean is the *lower* of the two — small *here*, because the last batch is 16 of 384 and the model is not much better or worse on it; the sign says only that those 16 went worse than the rest. Change either and it grows: with batch size 3,000 on 10,000 images the short batch carries a quarter of the weight instead of a tenth. `drop_last=False` is the default and that is the right default. The bug is in how the metric is aggregated, not in the loader.

Weight by the batch size, or count over the set — never take a plain mean of per-batch metrics. Not because the error is always large, but because *you cannot tell from the number whether it was*.

A custom module subclasses `nn.Module` and writes `forward` itself. One rule: `super().__init__()` before anything else. It creates the registry the layers and parameters are recorded in, so assigning a layer before it fails at once with `AttributeError: cannot assign module before Module.__init__() call`; everything after that — `.to(device)`, `.parameters()`, `state_dict()` — comes for free.

6.1 Searching the hyperparameters properly

Lecture 7 tuned by hand: ten points, one axis at a time. **Optuna** proposes a configuration, trains it briefly, scores it, and lets a sampler choose the next from what it has learned — here a learning rate on a log scale between 10^{-4} and 10^{-2} and a dropout rate in steps of 0.1. It is a search over configurations, not a final model. Not examinable: Optuna is not in the book. It is here because it is what you will actually use, and because it makes the point that hand tuning was never a search.

6.2 Saving the model, and checking the reload

```
torch.save(model.state_dict(), "sorter.pt")      # the tensors
fresh = build_model().to(device)
fresh.load_state_dict(torch.load("sorter.pt", weights_only=True))
```

The tensors are 1041 KB, 266,610 float32 numbers; the file weighs a little more, because `state_dict` is written as a zip archive with the names and the container inside it. Do not `torch.save(model, ...)`: that pickles the object, and unpickling needs your class definition at the same import path, so next term, when the file has moved, the checkpoint will not load.

Then check the reload, with an exact equality:

```
assert accuracy(model, X_test, y_test) == accuracy(fresh, X_test, y_test)
```

Here `accuracy` counts hits over the whole set, in `eval()` mode, under `no_grad`. Not "it looks about the same": a reloaded network is either the same function or it is not. The usual cause of failure is loading a `state_dict` into a model built with different widths. PyTorch raises for that — but only if you did not pass `strict=False` to silence it.

7 Where we ended

Model	Validation accuracy
Always one class	10.0%
Scikit-Learn MLP, CPU	87.38%
PyTorch, same model, MPS	86.32%
The same code without `zero_grad()`	20.44%

The last row is the one to remember: it ran, it finished, and it reported a number. The PyTorch model's test accuracy, touched once, is 85.35%.

Silent failure	Diagnosed in	Measured cost
Scaler fitted before the split	Part I	nothing measurable
Evaluating on training data	Part I	an RMSE of zero
Accuracy under class imbalance	Part I	a useless model above 90%
Hyperparameters chosen on the test set	Part I	an optimistic report
Missing `optimizer.zero_grad()`	Lecture 8	65.88 points
Missing `model.eval()`	Lecture 8	0.79 points
Metric averaged per batch	Lecture 8	0.15 points

Every one of them ran to completion and printed a plausible number.

8 The notebook

What to change

Make the network four times deeper and re-run the training loop, as the notebook's closing cell suggests. It trains far worse than its size suggests it should — and Lecture 9 derives why, and repairs it.
Either device is fine: everything is sized for the free CPU tier, and §3 shows how little an accelerator buys at this size.

9 Design exercise: the training pipeline behind a ride-hailing app's arrival-time model

The lecture closes with a design exercise in the manner of an ML system-design interview, on the system behind a company's product and on the thing this lecture bought: the loop, owned.

The task. A ride-hailing app shows every rider, at booking, how many minutes away the driver is. The number comes from a network trained on past trips, which a team of several engineers retrains every week as cities and traffic change. An estimate that runs early leaves a rider waiting and cancelling; one that runs late loses the booking. The pipeline must take a configuration and produce a checkpoint, a validation error and a record of how both were made, such that a colleague can reproduce the number and a reviewer can release it. Given: the trip log — request time, pickup, the driver's position at dispatch, traffic, and when the driver actually arrived — the model class, and a training loop written out in full. Millions of trips a week across many cities, a model that fits on one machine, many runs a day across the team. The output feeds the engineer who retrains it, the reviewer who releases it, and the service that loads the checkpoint for every booking.

1. **Frame it honestly.** A run outputs three things: the checkpoint, the mean absolute error in minutes counted over the whole validation set, and the configuration and seed that produced them. The baselines are the rule the app shipped first — distance at the city's average speed — and last week's checkpoint, scored by the same function on the same split; the reference is the same network under the convenience wrapper, whose training-loss curve the loop must overlay, the first check after any rewrite. *Pitfall:* every failure in this lecture ran to completion

and printed a plausible number, and a release gate that looks for exceptions passes all of them into every booking in every city. Judge runs on behaviour: the same configuration twice must give the same curve, the loop's curve must sit on the wrapper's, and the new checkpoint must beat last week's on the same split, or it is not released.

2. **`Dataset`, `DataLoader`, and a split written to disk.** A `Dataset` answers how many and give me trip *i*; a `DataLoader` shuffles and batches, and the interface does not change when a week of trips stops fitting in memory. Split once per retrain, by time — the latest days are the validation set — and save the indices, so every engineer's validation set is the same file; shuffle the training loader with a seeded `Generator`, and do not shuffle the validation loader. *Pitfall:* `shuffle=True` with no generator means every run sees the batches in a different order, and two engineers comparing a tenth of a minute are comparing shuffles; a split redone from another seed on another machine moves trips between training and validation, and the number is of a different experiment. One split file, one generator, both named in the configuration.

3. **The loop, written out.** `model.train()`, then per batch `zero_grad`, forward, loss, `backward`, `step` — five lines, every one a place to print, clip, log or stop. The loss is the one the product asked for, the absolute error in minutes, which the wrapper had no argument for; one loop function for the whole team, reviewed once, with the per-epoch loss logged on every run. *Pitfall:* `backward()` adds to `p.grad`, so a loop without the clearing line steps on the sum of every gradient so far; it runs, the loss falls for a while, and the model it produces is little better than the rule it was meant to replace, with no exception, no warning and no `NaN`. It is caught by reading the loop — which is why the team has one loop and not one per engineer — and by the overlay against the wrapper's curve.

4. **One `evaluate()` for everyone.** `model.eval()` first, or dropout stays on and the error is of a randomly crippled network; under `torch.no_grad()`, or a graph is recorded for nothing; sum the absolute errors and divide by the trip count, or a plain mean of per-batch errors weights the short last batch as a full one; and run it twice, because an error that wobbles between identical calls has a layer still in training mode. *Pitfall:* each engineer writes their own evaluation and each omits a different line. The missing `eval()` cost under a point today and the per-batch mean a small fraction of one; neither can be seen in the number, and either decides a release. One function, called by the loop, the search, the reload check and the release gate alike.

5. **Seeds.** Seed everything that draws a random number — the weights, the loader's generator, the split — and record the seed with the configuration; the same seed on the same machine must give the same curve. A difference between two configurations is a claim only when it is larger than the spread between seeds of one configuration, so run a few seeds and report mean and spread. *Pitfall:* an engineer reports a gain of a few seconds of error from a new dropout rate, measured on one seed against a colleague's one seed, when the seeds alone move the number by as much, and the release meeting argues about a change that did nothing. Reproduce first, repeat second, then compare.

6. **A hyperparameter search with a budget.** Hand tuning was never a search. An Optuna `objective` proposes a configuration, trains it for a few epochs, scores it with the team's `evaluate()` on the validation split, and the sampler chooses the next; fix the number of trials and the epochs per trial in advance, so the search fits inside the week's retrain, log every trial, retrain the winner with the full budget, and read the test split — the last day, held back —

once. *Pitfall:* the best trial's validation error reported as the model's. It is the minimum of many draws on the set it was chosen on, so it is optimistic, and it was earned on a short budget, so the ranking is a measurement of that budget. The search chooses a configuration; the number comes from the retrained winner, on trips the search never saw.

7. **Save the tensors, and check the reload.** `torch.save(model.state_dict(), path)` — the tensors and their names, nothing else — beside the configuration that built the model, so that the serving service can call `build_model(config)` without importing any of the training code; reload with `weights_only=True` into a freshly built model and assert that `evaluate()` returns exactly the same error, in the pipeline and again in the service when it loads. *Pitfall:* `torch.save(model, path)` pickles the object, and unpickling needs the class at the same import path — the service has a different one, and the checkpoint fails to load at release; and `strict=False` loads a `state_dict` into a model of different widths in silence, with layers missing, so every rider gets an estimate from a model nobody validated. A reloaded network is either the same function or it is not; the difference is one assertion, and it runs in the pipeline.

8. **In production: the weekly release, the rollout, and the clock.** A checkpoint that passes the gate goes to a share of bookings in one city against last week's, read on the error against real arrivals and on cancellations — the A/B test; last week's file is the rollback, and the engineer who retrained it is on call. Wall-clock is a measurement of one machine: time pairs of runs that differ in one thing — the wrapper against the loop on the same CPU is the framework, the loop on the CPU against the loop on the accelerator is the device — and record the machine, the device and the batch size with every timing. *Pitfall:* "PyTorch on the GPU is several times faster" — one number for two changes, read on one laptop, written into a ticket that buys hardware for the weekly retrain. Taken one at a time, the device alone nearly ties at the smallest width. A second is a reading from a named machine, and the pipeline stores it as one.

9. **Ship it.** Configuration and seed → the split file, by time → seeded loaders → the one loop, overlaid on the wrapper's curve → the one `evaluate()`, run twice → a few seeds before any comparison → a budgeted search, the winner retrained → `state_dict` with its configuration, the reload asserted in the pipeline and the service → timings with their machine → the gate, one city, the A/B test, the rollback ready.

10 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 9's deck and in the exercise book.

1. Reverse-mode automatic differentiation costs one sweep for all partial derivatives; forward mode costs one per input. State the condition on the function's shape that makes reverse mode the right choice. [4]

2. Test accuracy on 10,000 images is computed as the plain mean of the per-batch accuracies, in batches of 384. State why this is wrong, and the rule that fixes it. [3]

3. The five lines of the training loop are reordered so that `opt.step()` comes before `loss.backward()`. State what the student observes. [4]

4. A network with dropout is evaluated without calling `model.eval()`. Describe the symptom precisely, and the one-line diagnostic that catches it. [5]

5. In the measured comparison the accelerator barely beat the CPU on the smallest network, and its lead grew as the layers widened. Give the reason, and the quantity that decides where the crossover sits. [4]

Summary

The chain rule is a product of Jacobians and the product may be bracketed from either end. Forward mode costs one sweep per input; reverse mode costs one sweep per output; training has 266,610 inputs and one scalar output, so reverse mode is the only viable choice — by a factor equal to the parameter count. Backpropagation is that reverse sweep, seeded with 1 at the loss.

A cost in time is read in pairs, one change at a time; the accelerator's lead grows with the arithmetic each batch carries, from near-parity at the smallest width. The rebuilt model is the *same* model, and what was bought is the five-line loop.

Then three bugs that raise nothing, warn nothing, and return a plausible number. A missing `zero_grad()` costs 65.88 accuracy points because `backward()` accumulates by design; a missing `model.eval()` costs 0.79, and its tell is a 0.44-point spread across ten identical calls; a plain mean of per-batch accuracies is off by 0.15 points here, by an amount the number itself cannot reveal.

The closing design exercise built the training pipeline behind a ride-hailing app's arrival-time model in nine steps — runs judged on behaviour, one split file by time and seeded loaders, one reviewed loop, one `evaluate()` run twice, seeds repeated before a claim, a budgeted search with the winner retrained, a checkpoint asserted on reload in the service too, and a weekly release with an A/B test and a rollback — each step named for the pitfall it avoids.